# Hospital Chatbot for Color-Coded Clinical Pathways

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# Contents

# From chat and vitals to one triage colour

**Today's question:** How does the system turn what a patient says and what we know about their vitals into **one** urgency colour?

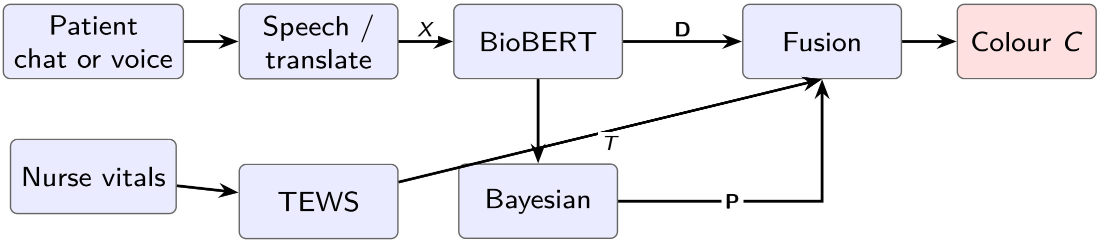
**This talk:** The **mathematics** behind that decision — three layers (language, vitals, probability) and a fusion step — not a repeat of the general project overview.

**The colour** $C$ follows **SATS** (South African Triage Scale):

- Red — immediate, life-threatening
- Orange — very urgent
- Yellow — urgent
- Green — non-urgent

The system supports triage staff; it does **not** diagnose disease or replace clinical judgment.

# How three mathematical layers produce one triage colour




| | |
|---|---|
| $X$ | English symptom text from the chatbot (after speech/translation) |
| **D** | Language-layer scores: which SATS danger patterns BioBERT detected |
| $T$ | TEWS total from nurse or reported vitals (whole number) |
| **P** | Probabilities of each triage colour when vitals are incomplete |
| $C$ | Final SATS triage colour assigned to the patient |

# Why the pipeline flows this way

- **Top path (words first):** Symptoms arrive as chat or voice. Translation comes *before* BioBERT because the model only reads medical English — we must normalise language before any NLP runs.
- **Bottom path (vitals in parallel):** Pulse, breathing, and temperature follow a separate branch because they are *numbers*, not sentences. TEWS turns them into a standard hospital score $T$ that does not depend on word choice.
- **Why BioBERT feeds fusion directly:** Language can flag SATS danger (e.g. crushing chest pain) even when the vital-score path looks mild — so text must reach fusion without waiting for a nurse.
- **Why Bayesian sits under BioBERT:** When vitals are missing or self-reported only, algebra alone is blind. Bayesian uses the same symptom evidence plus whatever numbers exist to estimate risk — it runs when TEWS is incomplete.
- **Why fusion is last:** No single layer is trusted alone. Fusion applies priority rules so a strong symptom signal or a high $T$ cannot be ignored, and outputs one final colour $C$ for the hospital pathway.

# What fusion does: combining language, vitals, and probability safely

**Fusion** is the **master controller** function $f_{\text{fusion}}$: it reads three outputs and assigns the final colour $C$.

- **D** — danger patterns found in text (from BioBERT)
- $T$ — TEWS score from measured vitals (whole number)
- **P** — probabilities of each colour when evidence is incomplete

**Safety rule:** Fusion never assigns a **less urgent** colour than a strong language signal or a high $T$ would require. It prevents under-triage when vitals alone look mild.

# How fusion works: the mathematics

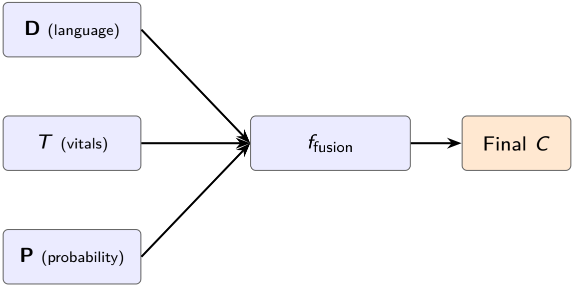
**In plain terms:** Fusion is the **decision step**. It does *not* mix $\mathbf{D}$, $T$, and $\mathbf{P}$ into one blended score. It runs a **hospital-style checklist** — top to bottom — and stops at the **first serious finding**.

**What it receives:**

- From words: did BioBERT flag a dangerous SATS pattern? ($\mathbf{D}$)
- From vitals: what is the TEWS total? ($T$)
- From probability: if vitals are incomplete, which colour is most likely? ($\mathbf{P}$)

**How it decides $C$:** Check (1) life-threatening words → Red; else (2) very urgent words → Orange; else (3)–(5) use $T$ bands; else if data missing use the highest-probability colour from $\mathbf{P}$; else Green.
Shorthand: $C = f_{\text{fusion}}(\mathbf{D}, T, \mathbf{P})$.

# Fusion inputs and outputs




Urgency order: $\mathrm{ord}(\text{Red}) = 4 > \mathrm{ord}(\text{Orange}) = 3 > \mathrm{ord}(\text{Yellow}) = 2 > \mathrm{ord}(\text{Green}) = 1$. When layers disagree, fusion picks the safer (higher) urgency.

# How patients report symptoms in the chatbot

Users type in the **chat window** or send **voice notes**; both become symptom text $X$ for the NLP layer.

**Typed chat:** The patient message is stored as $X$ (symptom text in English for BioBERT).
**Voice:** Sound is converted to text, then to English:

$$X = g_{\text{trans}}(g_{\text{ASR}}(\mathrm{audio}))$$

Here $g_{\text{ASR}}$ = speech-to-text (sound $\rightarrow$ letters) and $g_{\text{trans}}$ = machine translation (e.g. Twi $\rightarrow$ English).

One patient visit is tracked as session $s$ so all modules share the same record.

# Why medical English is required for BioBERT

BioBERT was trained on biomedical **English** (PubMed, clinical notes). The model expects input $X$ in English.

- Twi or other local languages are translated before NLP runs.
- Google Cloud Translation offers a free tier ($\sim$500k characters/month); **Twi may not be supported** on the API — a glossary fallback may be needed.

Translation quality affects which danger words reach BioBERT; clinical review of common phrases is part of deployment.

# Turning words into vectors the model can compare

The tokeniser $\tau$ splits $X$ into pieces $t_1, \ldots, t_M$. Each piece gets a vector:

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(t_i)$$

$E_{\text{word}}$ = meaning of the piece; $E_{\text{pos}}(i)$ = position in the sentence; $d = 768$ is the vector size (BERT-Base). Token `[CLS]` at the start will summarise the whole message.

# Letting each word attend to the rest of the sentence

Each Transformer layer uses **self-attention** so words share context:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

$Q$ (query), $K$ (key), and $V$ (value) are learned projections; $d_k$ scales the scores; softmax turns them into weights summing to 1. The stack has $L = 12$ layers and $H = 12$ parallel heads per layer.

# Detecting SATS danger patterns in the text

The NLP function $f_{\text{NLP}}$ maps text $X$ to discriminator scores:

$$\mathbf{D} = f_{\text{NLP}}(X;\, \theta_{\text{BERT}}), \qquad d_j = \sigma(w_j^\top h_{\texttt{[CLS]}} + b_j) \in [0, 1]$$

Each $d_j$ is confidence that SATS discriminator $j$ is present (e.g. central chest pain). If $d_j > \tau$ (e.g. $\tau = 0.85$), that pattern counts as detected. NER = locating medical terms in $X$.

# Scoring vital signs with the TEWS sum

**TEWS** (Triage Early Warning Score) is the deterministic function $f_{\text{TEWS}}$:

$$T = f_{\text{TEWS}}(\mathbf{v}) = \sum_{k=1}^{6} w_k\, f_k(v_k), \quad w_k = 1$$

Vitals $\mathbf{v}$: $v_1$ mobility, $v_2$ respiratory rate (RR), $v_3$ heart rate (HR), $v_4$ temperature, $v_5$ AVPU (consciousness), $v_6$ trauma. Each $f_k(v_k)$ returns 0–3 points from SATS tables.

# Mapping the TEWS total to a colour band

Example: heart rate scoring $f_1$ when $v_1 = \mathrm{HR}$:

$$f_1 = \begin{cases} 3 & \mathrm{HR} \geq 130 \\ 2 & 111 \leq \mathrm{HR} \leq 129 \\ 0 & 51 \leq \mathrm{HR} \leq 100 \\ 2 & \mathrm{HR} \leq 40 \end{cases}$$

Colour from vitals alone, $C_{\mathsf{TEWS}}(T)$: $\begin{array}{ll} T > 7 & \text{Red} \\ 5 \leq T \leq 6 & \text{Orange} \\ 3 \leq T \leq 4 & \text{Yellow} \\ T \leq 2 & \text{Green} \end{array}$

# Estimating colour probability when evidence is partial

When vitals are missing, $f_{\text{Bayes}}$ updates beliefs:

$$P(C_k \mid E) = \frac{P(E \mid C_k)\, P(C_k)}{\sum_j P(E \mid C_j)\, P(C_j)}$$

$C_k$ = colour class $k$; $P(C_k)$ = **prior** (how common $k$ is at the hospital); $P(E \mid C_k)$ = **likelihood** (how often evidence $E$ appears if colour is $k$).

# Building evidence from symptoms and measured vitals

Evidence bundles language and vitals:

$$E = \mathrm{concat}(\text{symptoms from NLP},\ \mathbf{v}_{\text{obs}})$$

If some vitals are missing: $T_{\text{partial}} = \sum_{k \in \text{observed}} w_k f_k(v_k)$, and fusion may use $C = \arg\max_k P(C_k \mid E)$ (pick the colour with highest posterior probability).

# Applying the fusion algorithm in priority order

**Same rules as the Fusion section** (slide 7). $f_{\text{fusion}}$ checks in order (first match wins):

1. Red discriminator in $\mathbf{D}$ $(d_j > \tau) \Rightarrow C =$ Red
2. Orange discriminator in $\mathbf{D} \Rightarrow C =$ Orange
3. $T > 7 \Rightarrow C =$ Red
4. $5 \leq T \leq 6 \Rightarrow C =$ Orange
5. $3 \leq T \leq 4 \Rightarrow C =$ Yellow
6. Vitals incomplete $\Rightarrow C = \arg\max_k P(C_k \mid E)$
7. Else $C =$ Green

# Resolving conflicts so patients are not under-triaged

| **Language D** | **TEWS** $T$ | **Final** $C$ |
|---|---|---|
| High (chest pain) | $T = 4$ (Yellow alone) | Orange |
| Low | $T \geq 7$ | Red |

A strong discriminator overrides a mild $T$. High $T$ overrides weak language. Fusion always errs toward the more urgent colour when signals conflict.

# Data and software required to run the system

| **Needed** | **Purpose** |
|---|---|
| Patient text or audio | Input $X$ |
| SATS tables $f_k$, $w_k$ | Compute $T$ |
| Discriminator list | Labels for $\mathbf{D}$ |
| Prior / likelihood tables | Bayesian $\mathbf{P}$ |
| BioBERT weights $\theta_{\text{BERT}}$ | Run $f_{\text{NLP}}$ |
| Session store | All layers share one record $s$ |

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